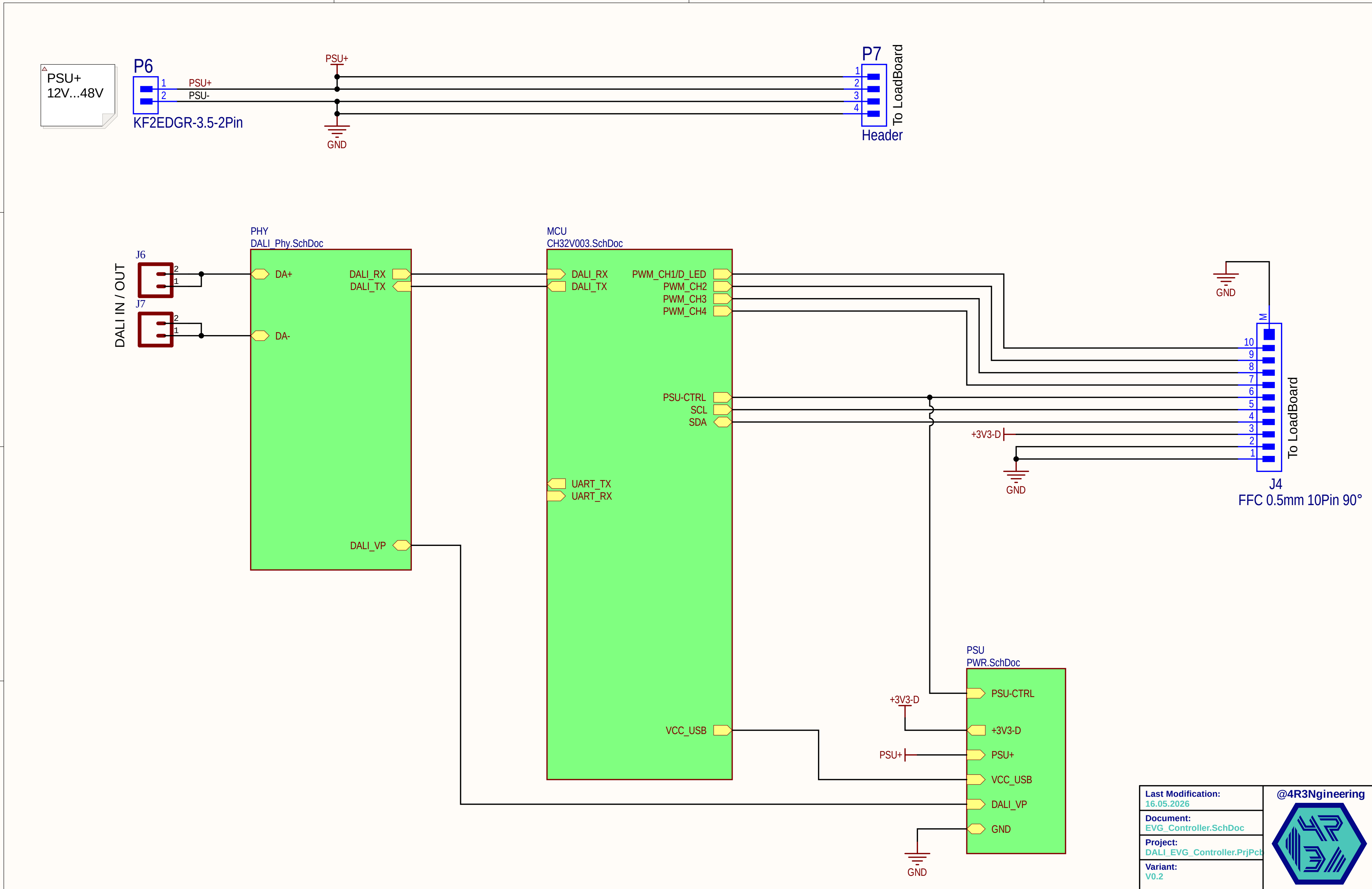
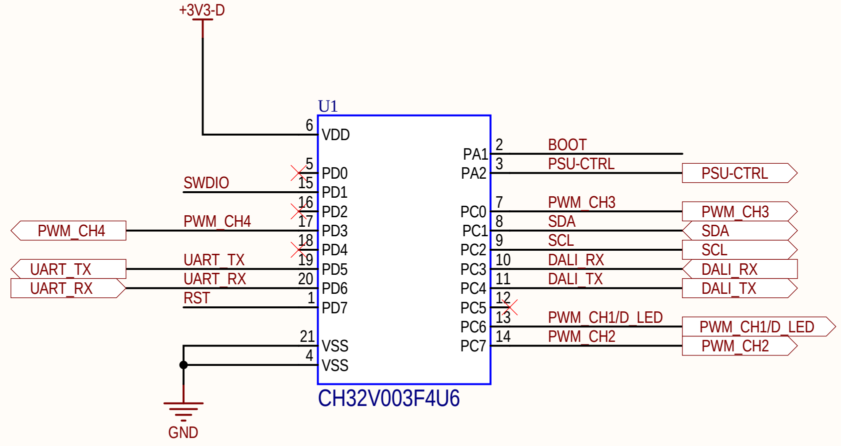
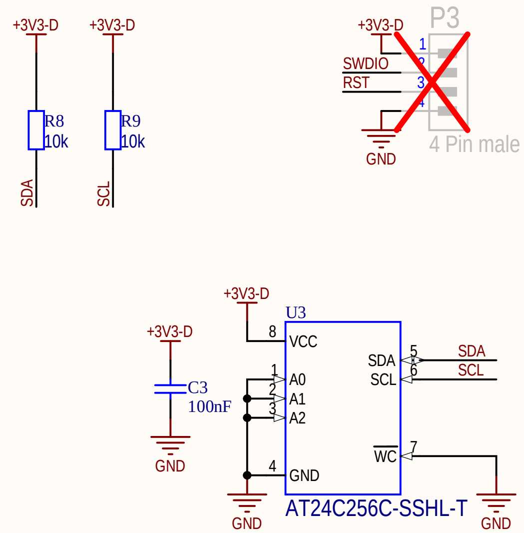


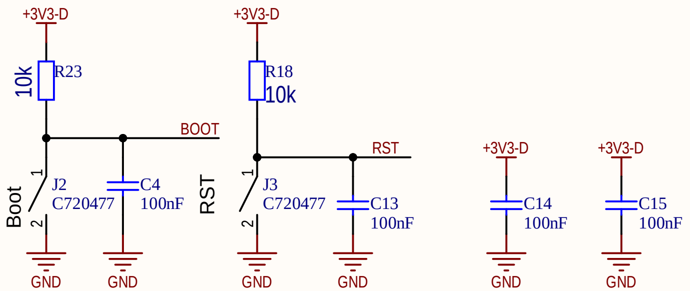
Last Modification: 16.05.2026	@4R3Ngineering
Document: EVG_Controller.SchDoc	
Project: DALI_EVG_Controller.PrjPcb	
Variant: V0.2	

USB HID Bootloader from CNLohr Compiled with:

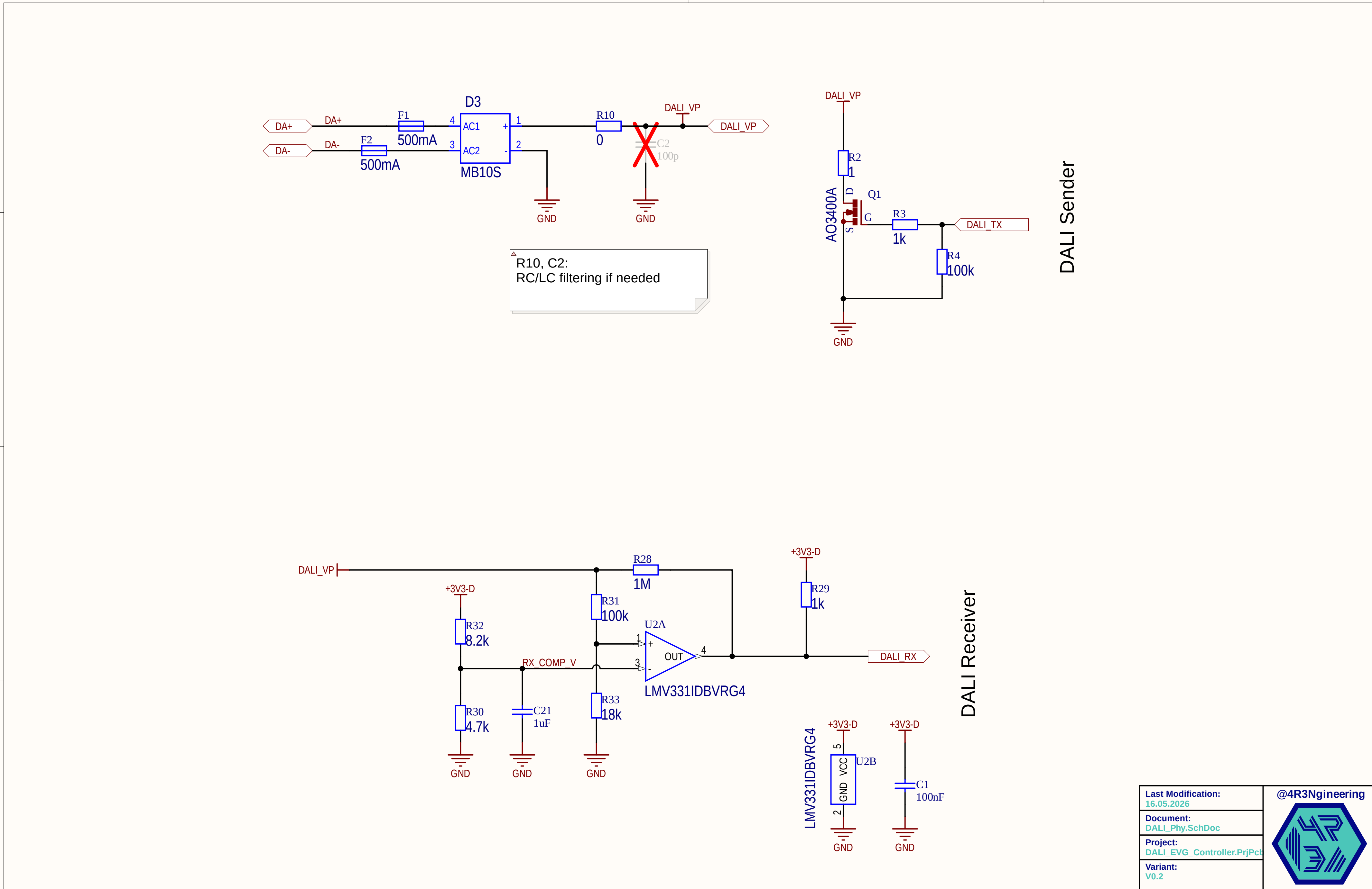
```
<bootloader.c>
#define DISABLE_BOOTLOAD

#define BOOTLOADER_BTN_PORT D
#define BOOTLOADER_BTN_PIN 0
#define BOOTLOADER_BTN_TRIG_LEVEL 0

#define BOOTLOADER_TIMEOUT_PWR 0

<usb_config.h>
#define USB_PORT D
#define USB_PIN_DP 4
#define USB_PIN_DM 3

#define USB_PIN_DPU 0
```

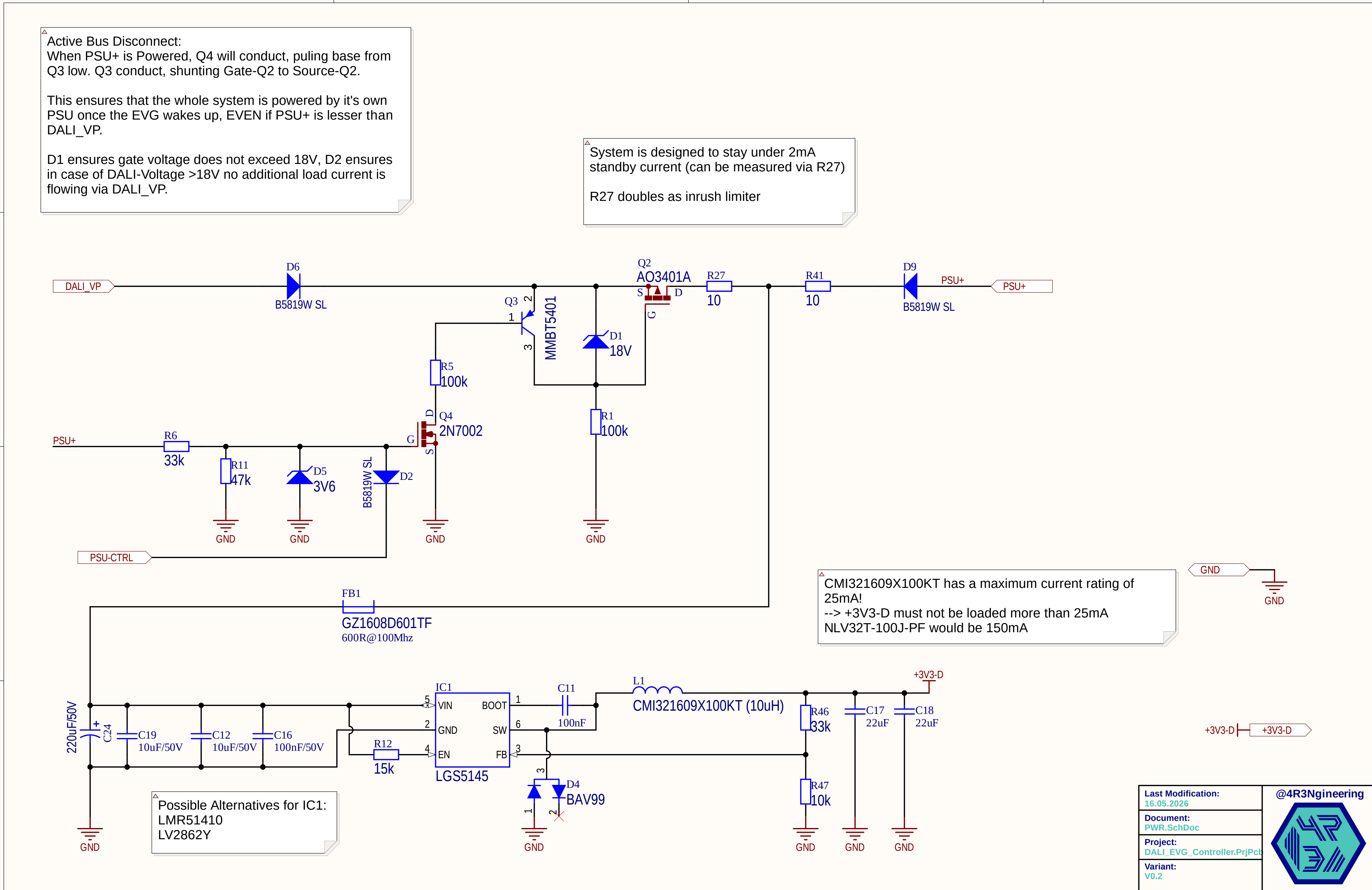






Last Modification: 16.05.2026	@4R3Ngineering
Document: CH32V003.SchDoc	
Project: DALI_EVG_Controller.PrjPcb	
Variant: V0.2	


DA+
DA-
F1
500mA
F2
500mA
D3
AC1
AC2
MB10S
R10
0
C2
100p
DALI_VP
GND
R10, C2:
RC/LC filtering if needed
R2
1
Q1
AO3400A
D
G
S
R3
1k
DALI_TX
R4
100k
DALI Sender
R28
1M
R31
100k
+3V3-D
R32
8.2k
RX_COMP_V
U2A
OUT
LMV331IDBVRG4
DALI_RX
R29
1k
R30
4.7k
C21
1uF
R33
18k
DALI Receiver
U2B
VCC
GND
LMV331IDBVRG4
C1
100nF
Last Modification:
16.05.2026
Document:
DALI_Phy.SchDoc
Project:
DALI_EVG_Controller.PrjPcb
Variant:
V0.2
@4R3Ngineering


Active Bus Disconnect:
When PSU+ is Powered, Q4 will conduct, puling base from Q3 low. Q3 conduct, shunting Gate-Q2 to Source-Q2.
This ensures that the whole system is powered by it's own PSU once the EVG wakes up, EVEN if PSU+ is lesser than DALI_VP.
D1 ensures gate voltage does not exceed 18V, D2 ensures in case of DALI-Voltage >18V no additional load current is flowing via DALI_VP.
System is designed to stay under 2mA standby current (can be measured via R27)
R27 doubles as inrush limiter
DALI_VP
D6
B5819W SL
Q3
MMBT5401
Q2
AO3401A
D1
18V
R27
10
R41
10
D9
B5819W SL
PSU+
R5
100k
Q4
2N7002
R1
100k
R6
33k
R11
47k
D5
3V6
D2
B5819W SL
GND
PSU-CTRL
FB1
GZ1608D601TF
600R@100Mhz
CMI321609X100KT has a maximum current rating of 25mA!
--> +3V3-D must not be loaded more than 25mA
NLV32T-100J-PF would be 150mA
IC1
VIN
GND
EN
BOOT
SW
FB
LGS5145
C11
100nF
L1
CMI321609X100KT (10uH)
+3V3-D
R46
33k
R47
10k
C17
22uF
C18
22uF
C24
220uF/50V
C19
10uF/50V
C12
10uF/50V
C16
100nF/50V
R12
15k
D4
BAV99
Possible Alternatives for IC1:
LMR51410
LV2862Y
Last Modification:
16.05.2026
Document:
PWR.SchDoc
Project:
DALI_EVG_Controller.PrjPcb
Variant:
V0.2
@4R3Ngineering